



Artificial Intelligence in Medicine:

Foundations and Clinical Applications

Instructor



Sara El-Sayed El-Metwally

Associate Professor of Computer Science,
Computer Science Department,
Faculty of Computers and Information,
Mansoura University



<https://sarael-metwally.github.io/SaraEl-MetwallyLab/>



sarah_almetwally4@mans.edu.eg



<https://www.facebook.com/sara.almetwally>

Google Teachable Machine

❑ Google Teachable Machine is a free, web-based tool that lets you train custom artificial intelligence models quickly without writing any code.

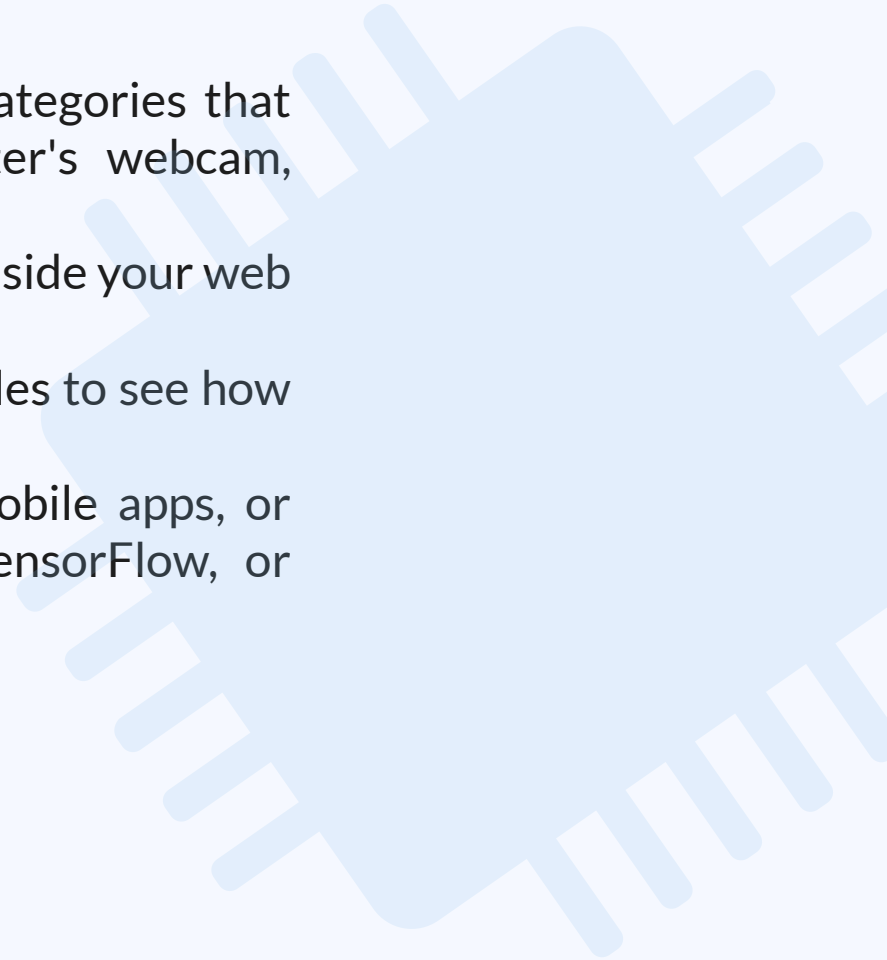
❑ How It Works

❑ **Gather:** Collect and group examples into different classes or categories that you want your computer to learn. You can use your computer's webcam, microphone, or upload image, audio, and pose files directly.

❑ **Train:** Click a button to train your machine learning model right inside your web browser. All data processing stays local and private on your device.

❑ **Test:** Instantly test your model using live inputs or new sample files to see how accurately it classifies your data with a confidence score.

❑ **Export:** Download your finished model for use in websites, mobile apps, or physical hardware projects using formats like TensorFlow.js, TensorFlow, or TensorFlow Lite.



<https://teachablemachine.withgoogle.com/>

Teachable Machine

Train a computer to recognize your own images, sounds, & poses.

A fast, easy way to create machine learning models for your sites, apps, and more – no expertise or coding required.

Get Started



<https://teachablemachine.withgoogle.com/>

New Project

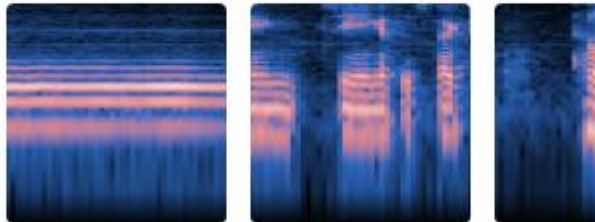
 Open an existing project from Drive.

 Open an existing project from a file.



Image Project

Teach based on images, from files or your webcam.



Audio Project

Teach based on one-second-long sounds, from files or your microphone.

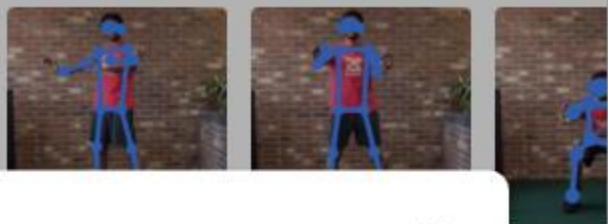
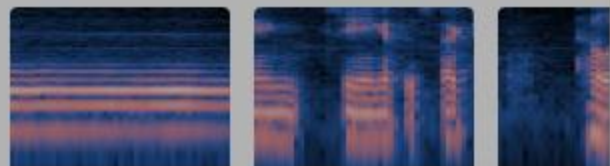


Pose Project

Teach based on images, from files or your webcam.

<https://teachablemachine.withgoogle.com/>

Teachable Machine



New Image Project

Standard image model

Best for most uses

224x224px color images

Export to TensorFlow, TFLite, and TF.js

Model size: around 5mb

Embedded image model

Best for microcontrollers

96x96px greyscale images

Export to TFLite for Microcontrollers, TFLite, and TF.js

Model size: around 500kb

[See what hardware supports these models.](#)

<https://teachablemachine.withgoogle.com/>

≡ Teachable Machine

Class 1 



Add Image Samples:



Webcam



Upload

Class 2 



Add Image Samples:



Webcam



Upload

Training

Train Model

Advanced



Preview



Export Model

You must train a model on the left before you can preview it here.

Google Teachable Machine: Gather

Skin Cancer or Not Skin Cancer Image Datasets



[Data Card](#) [Code \(4\)](#) [Discussion \(0\)](#) [Suggestions \(0\)](#)

About Dataset

No description available

Usability ⓘ
3.75

License
[MIT](#)

Expected update frequency
Not specified

<https://www.kaggle.com/datasets/mdismielhossenabir/skin-cancer-or-not-skin-cancer-image-datasets>

Google Teachable Machine: Gather

Skin Cancer or Not Skin Cancer Image Datasets



Data Card Code (4) Discussion (0) Suggestions (0)

Data Explorer

Version 1 (3.32 MB)

- ▼ test
 - test
 - not_skin_cancer
 - skin_cancer
- ▼ train
 - train
 - not_skin_cancer
 - skin_cancer

<https://www.kaggle.com/datasets/mdismielhossenabir/skin-cancer-or-not-skin-cancer-image-datasets>

Google Teachable Machine: **Edit Classes**

≡ Teachable Machine

Skin Cancer 

Add Image Samples:



Webcam



Upload

Not a Skin Cancer 

Add Image Samples:



Webcam



Upload

Training

Train Model

Advanced

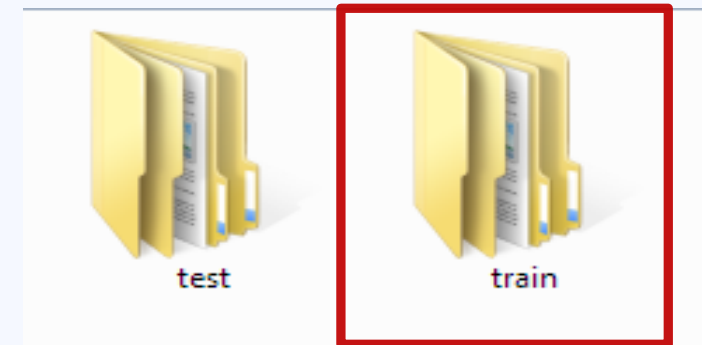
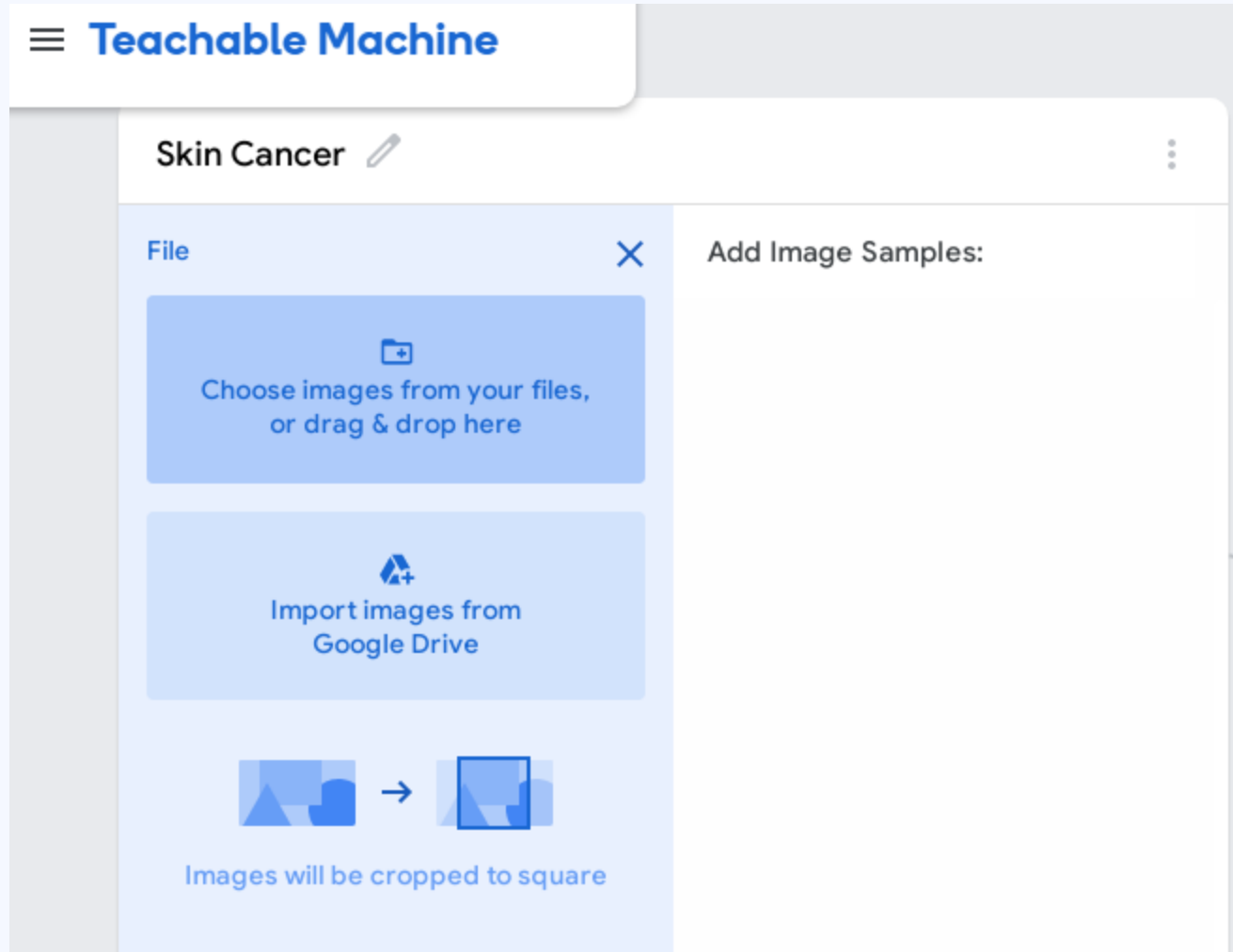
Preview



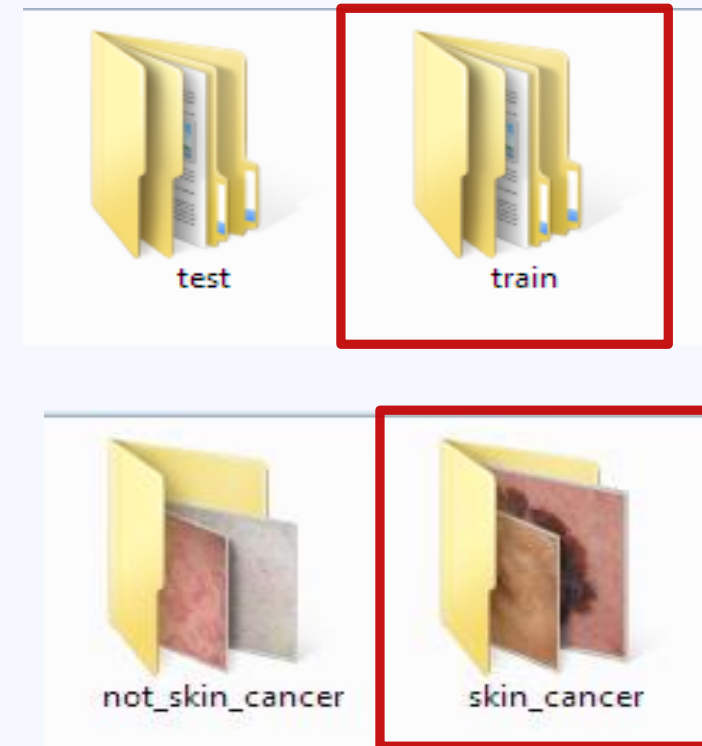
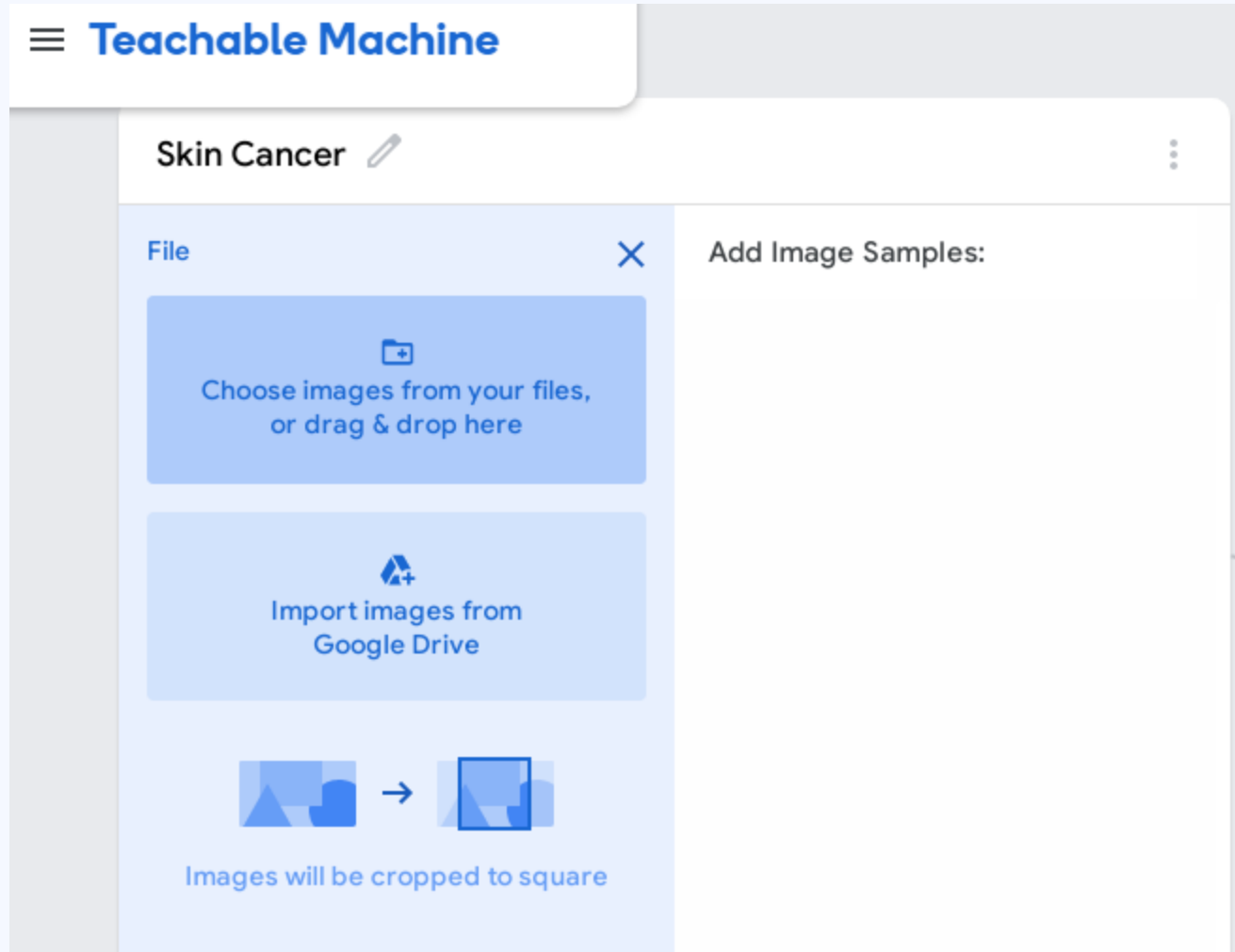
Export Model

You must train a model on the left before you can preview it here.

Google Teachable Machine: Image Upload



Google Teachable Machine: Image Upload



Google Teachable Machine: Image Upload of Skin Cancer

≡ Teachable Machine

Skin Cancer 

File 


Choose images from your files,
or drag & drop here


Import images from
Google Drive


Images will be cropped to square

117 Image Samples



Training

Train Model

Advanced 

Preview



Export Model

You must train a model on the left
before you can preview it here.

Google Teachable Machine: Image Upload of Not a Skin Cancer

The screenshot displays the Google Teachable Machine interface with two datasets. The 'Skin Cancer' dataset at the top has 117 image samples and includes 'Webcam' and 'Upload' buttons. Below it, the 'Not a Skin Cancer' dataset is highlighted with a red border; it has 93 image samples and a 'File' sidebar with options to choose images from files, import from Google Drive, and a note that images will be cropped to square. To the right, the 'Training' panel features a 'Train Model' button and an 'Advanced' dropdown. The 'Preview' panel on the far right contains an 'Export Model' button and a message: 'You must train a model on the left before you can preview it here.'

Google Teachable Machine: Model Training

Skin Cancer 

117 Image Samples

Webcam 

Upload 



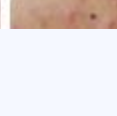
Not a Skin Cancer 

File 


Choose images from your files,
or drag & drop here


Import images from
Google Drive

93 Image Samples



Training

Train Model

Advanced 

Epochs: 50  

Batch Size: 16  

Learning Rate: 0.001  

Reset Defaults 

Under the hood 

Preview 

Export Model

You must train a model on the left before you can preview it here.

Google Teachable Machine: **Model Training**

Teachable Machine

Skin Cancer

117 Image Samples

Webcam

Upload



Not a Skin Cancer

File

Choose images from your files,
or drag & drop here

93 Image Samples



Training

Training...

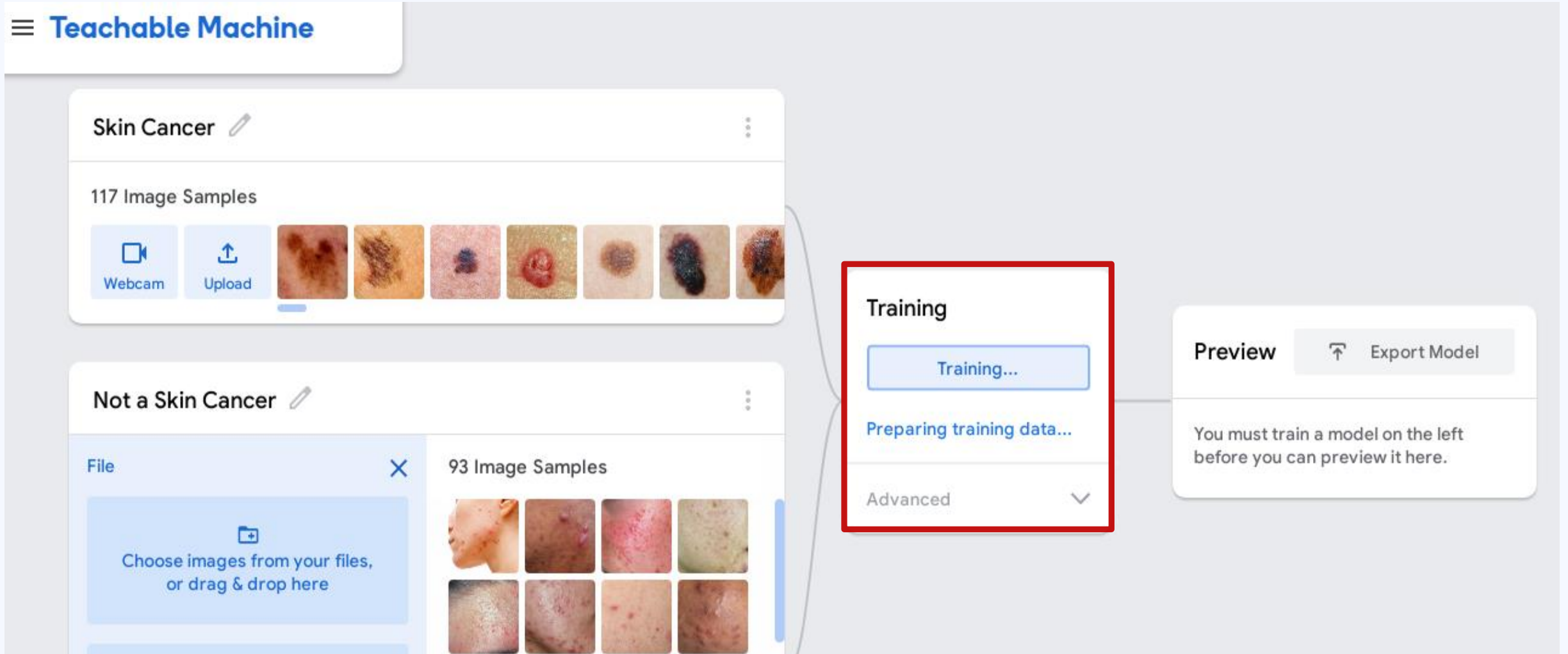
Preparing training data...

Advanced

Preview

Export Model

You must train a model on the left
before you can preview it here.



Google Teachable Machine: **Model Training**

≡ Teachable Machine

Don't switch tabs!
You must leave this tab open to train your model.

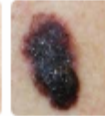
Don't show again OK

Skin Cancer ✎

117 Image Samples

Webcam

Upload



Not a Skin Cancer ✎

93 Image Samples

Webcam

Upload



Training

Training...

00:20 - 9 / 50

Advanced ▾

Preview

⤴ Export Model

You must train a model on the left before you can preview it here.

Google Teachable Machine: Model Testing

≡ Teachable Machine

Skin Cancer ✎

117 Image Samples



Webcam



Upload



Not a Skin Cancer ✎

93 Image Samples



Webcam



Upload



Training

Model Trained

Advanced



Preview



Export Model

Input



ON

Webcam



Webcam

File

There was an error opening your webcam. Make sure permissions are enabled or switch to image uploading.

Google Teachable Machine: Model Testing

Skin Cancer

117 Image Samples

Webcam

Upload



Not a Skin Cancer

93 Image Samples

Webcam

Upload



+

 Add a class

Training

Model Trained

Advanced

Input

ON

File

Choose images from your files, or drag & drop here

Import images from Google Drive

↓

Output

Skin Cancer

Not a Skin Cancer

Google Teachable Machine: **Model Testing**

Skin Cancer

117 Image Samples

Webcam

Upload

Not a Skin Cancer

93 Image Samples

Webcam

Upload

Training

Model Trained

Advanced

Input

ON

File

Choose images from your files, or drag & drop here

Import images from Google Drive

Output

Skin Cancer

Not a Skin Cancer

test

train

not_skin_cancer

skin_cancer

Google Teachable Machine: Model Testing

Skin Cancer

117 Image Samples

Webcam

Upload

Not a Skin Cancer

93 Image Samples

Webcam

Upload

Add a class

Training

Model Trained

Advanced

Input

ON

File

Choose images from your files, or drag & drop here

Import images from Google Drive

Output

Skin Cancer

Not a Skin Cancer

Google Teachable Machine: **Model Testing**

The screenshot displays the Google Teachable Machine interface during the model testing phase. On the left, two data classes are visible: "Skin Cancer" with 117 image samples and "Not a Skin Cancer" with 93 image samples. Each class has "Webcam" and "Upload" buttons. A "Training" panel in the center shows a "Model Trained" button and an "Advanced" dropdown. On the right, a red-bordered box highlights the testing interface. It includes an "Input" section with a toggle set to "ON" and a "File" dropdown. Below this are two options: "Choose images from your files, or drag & drop here" and "Import images from Google Drive". A large image of a skin lesion is shown in the center. At the bottom, the "Output" section displays two results: "Skin Cancer" with a 100% confidence score (represented by an orange bar) and "Not a Skin Cancer" with a lower confidence score (represented by a pink bar).

Skin Cancer ✎

117 Image Samples

Webcam Upload

Not a Skin Cancer ✎

93 Image Samples

Webcam Upload

⊞ Add a class

Training

Model Trained

Advanced ▾

Input ☒ ON File ▾

Choose images from your files, or drag & drop here

Import images from Google Drive



↓

Output

Skin Cancer 100%

Not a Skin Cancer

Google Teachable Machine: **Model Testing**

The screenshot displays the Google Teachable Machine interface during the model testing phase. On the left, the training data is organized into two classes: "Skin Cancer" with 117 image samples and "Not a Skin Cancer" with 93 image samples. Each class has "Webcam" and "Upload" buttons and a row of sample images. A "Training" panel in the center shows the status "Model Trained" and an "Advanced" dropdown. On the right, the "Model Testing" panel is highlighted with a red border. It features two options to choose images for testing: "Choose images from your files, or drag & drop here" and "Import images from Google Drive". Below these is a preview of the selected image, which shows a close-up of skin with red, inflamed patches. The "Output" section at the bottom of the testing panel displays the classification results: "Skin Cancer" with a low probability (represented by a short orange bar) and "Not a Skin Cancer" with a 100% probability (represented by a full red bar).

Skin Cancer 117 Image Samples

Webcam Upload

Not a Skin Cancer 93 Image Samples

Webcam Upload

⊕ Add a class

Training

Model Trained

Advanced

Choose images from your files, or drag & drop here

Import images from Google Drive

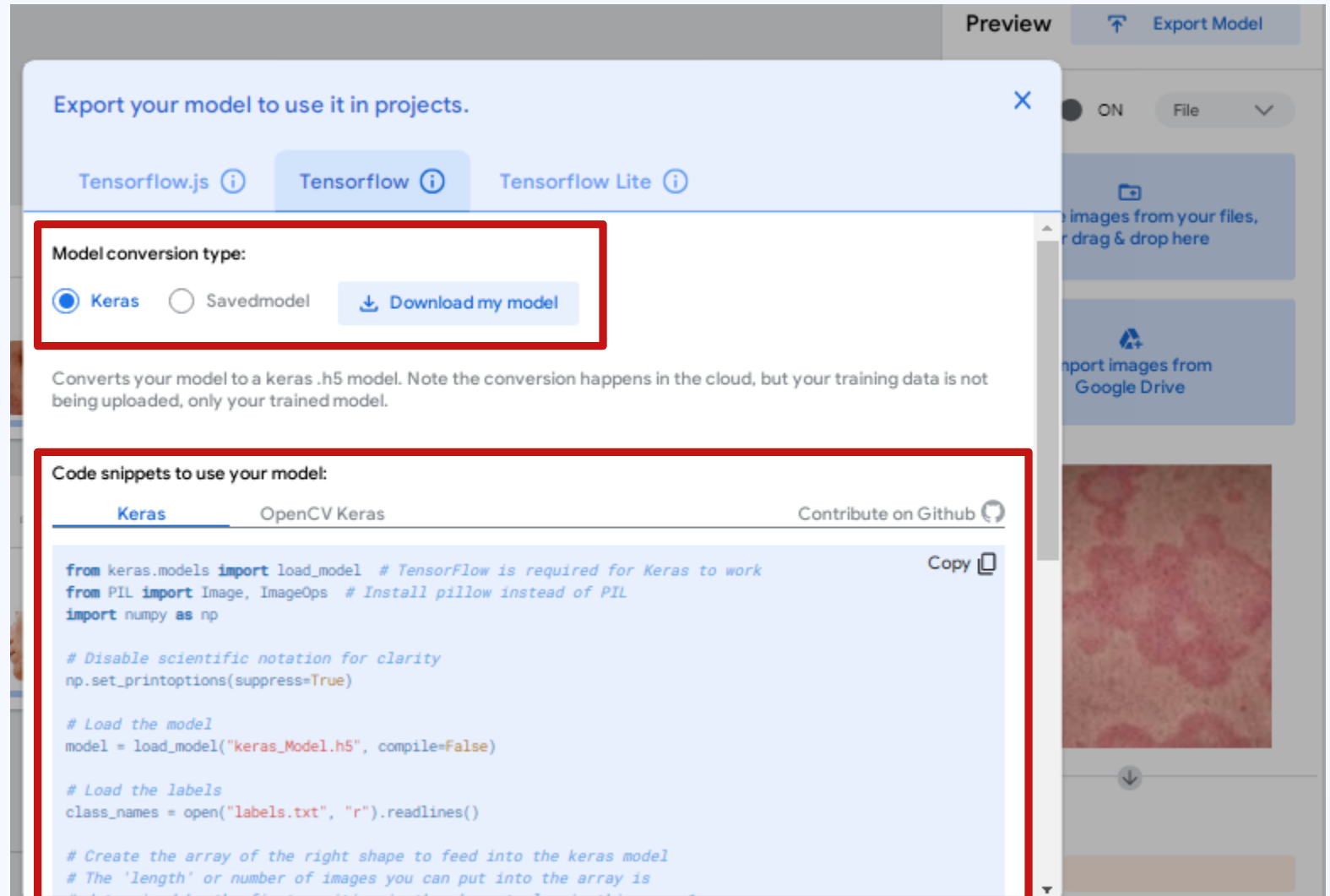
↓

Output

Skin Cancer

Not a Skin Cancer 100%

Google Teachable Machine: Model Export



Export your model to use it in projects.

Tensorflow.js ⓘ Tensorflow ⓘ Tensorflow Lite ⓘ

Model conversion type:

☒ Keras ☐ Savedmodel [Download my model](#)

Converts your model to a keras .h5 model. Note the conversion happens in the cloud, but your training data is not being uploaded, only your trained model.

Code snippets to use your model:

[Keras](#) [OpenCV Keras](#) [Contribute on Github](#)

```
from keras.models import load_model # TensorFlow is required for Keras to work
from PIL import Image, ImageOps # Install pillow instead of PIL
import numpy as np

# Disable scientific notation for clarity
np.set_printoptions(suppress=True)

# Load the model
model = load_model("keras_Model.h5", compile=False)

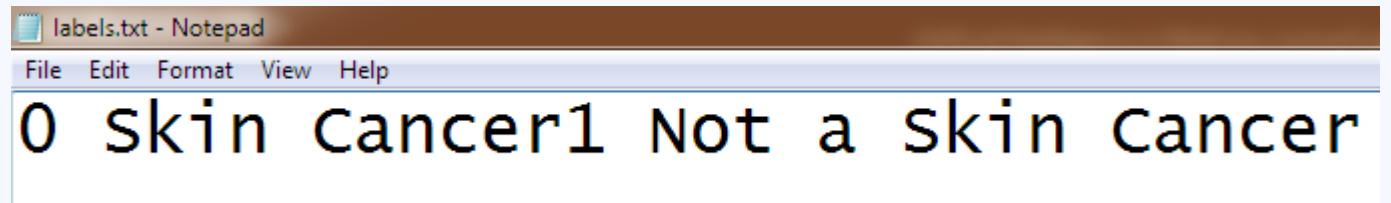
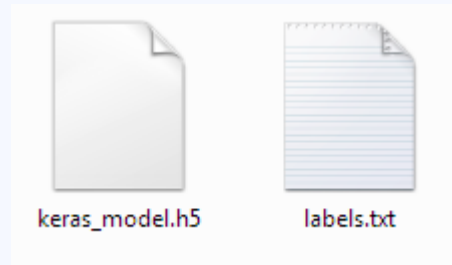
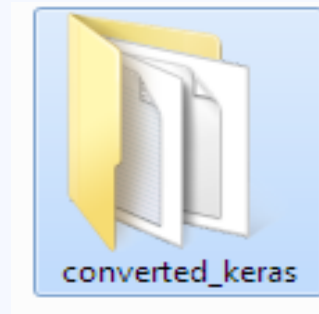
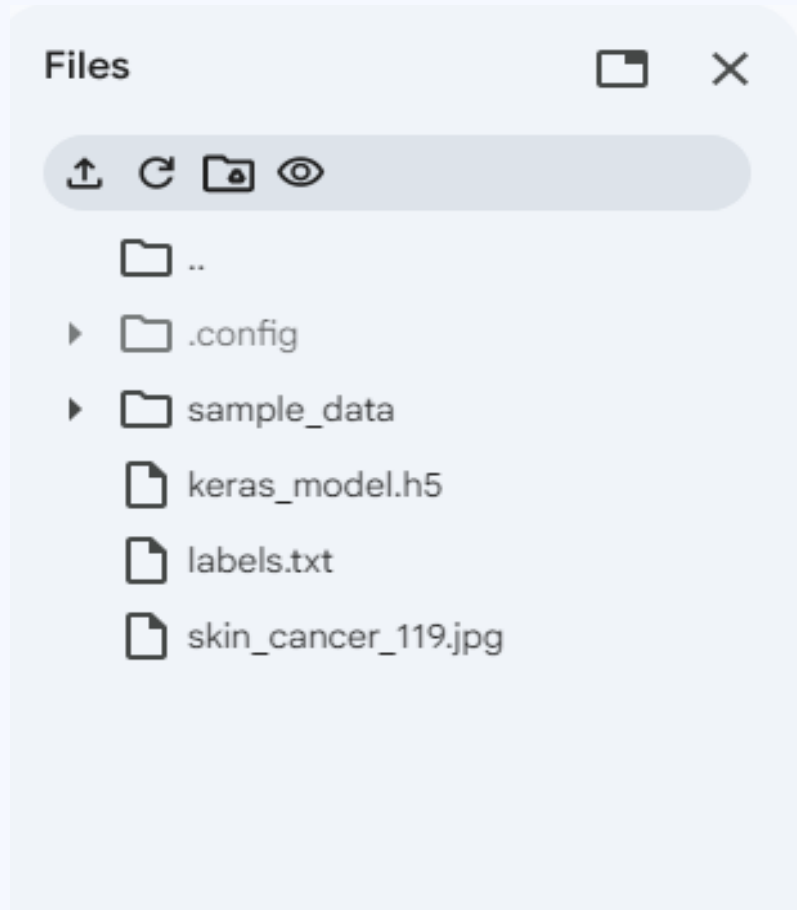
# Load the labels
class_names = open("labels.txt", "r").readlines()

# Create the array of the right shape to feed into the keras model
# The 'length' or number of images you can put into the array is
```

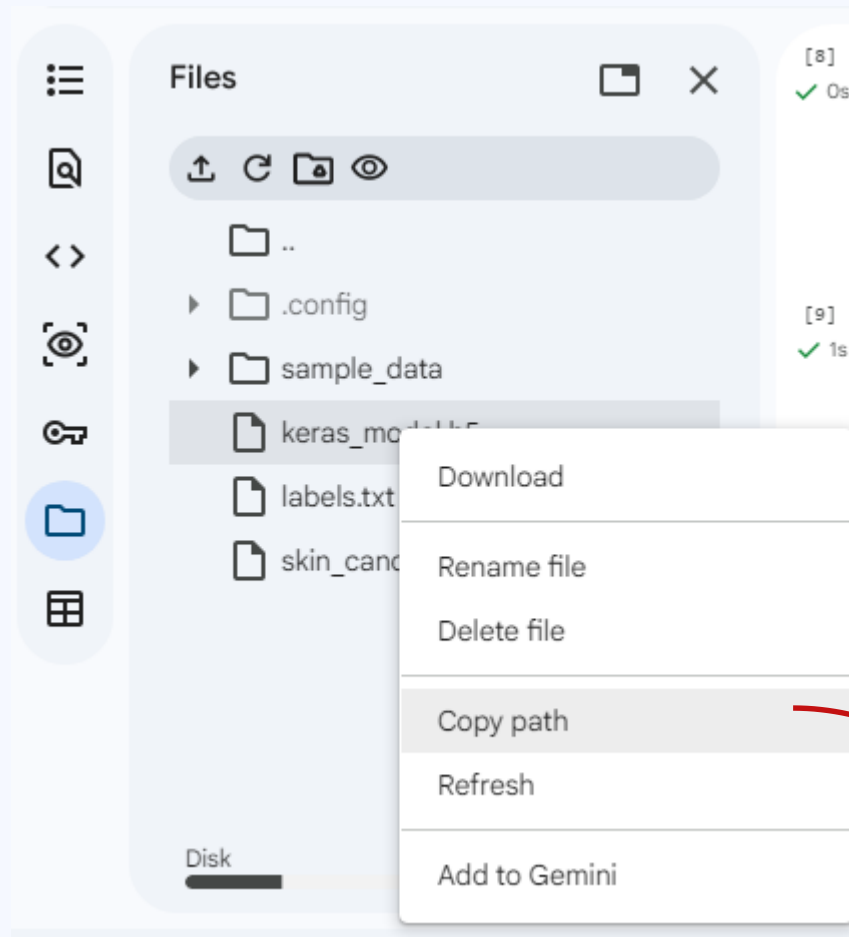
[Copy](#)

Converts your model to a keras .h5 model. Note the conversion happens in the cloud, but your training data is not being uploaded, only your trained model.

Google Teachable Machine: **Model Export**



Google Teachable Machine: **Model Export**



The model can be loaded after training to test it.

- **An H5 file (.h5)** is a legacy hierarchical data format used by Keras to save a complete deep learning model, including its architecture, trained weights, and optimizer state, into a single file.
- **.keras (Modern Native Keras Format)**, Default standard for Keras 3+.

```
model = tf.keras.models.load_model("/content/keras_model.h5", compile=False)
```

Google Teachable Machine: **Model Export**

```
from keras.models import load_model # TensorFlow is required for Keras to work
from PIL import Image, ImageOps # Install pillow instead of PIL
import numpy as np
```

```
import os
# Set Keras backend to legacy before importing tensorflow
os.environ["TF_USE_LEGACY_KERAS"] = "1"

import tensorflow as tf

model = tf.keras.models.load_model("/content/keras_model.h5", compile=False)
```

```
... Model loaded successfully!
```

```
# Load the labels
class_names = open("/content/labels.txt", "r").readlines()
```

Google Teachable Machine: **Model Export**

```
# Create the array of the right shape to feed into the keras model
# The 'length' or number of images you can put into the array is
# determined by the first position in the shape tuple, in this case 1
data = np.ndarray(shape=(1, 224, 224, 3), dtype=np.float32)

# Replace this with the path to your image
image = Image.open("/content/skin_cancer_119.jpg").convert("RGB")

# resizing the image to be at least 224x224 and then cropping from the center
size = (224, 224)
image = ImageOps.fit(image, size, Image.Resampling.LANCZOS)

# turn the image into a numpy array
image_array = np.asarray(image)

# Normalize the image
normalized_image_array = (image_array.astype(np.float32) / 127.5) - 1

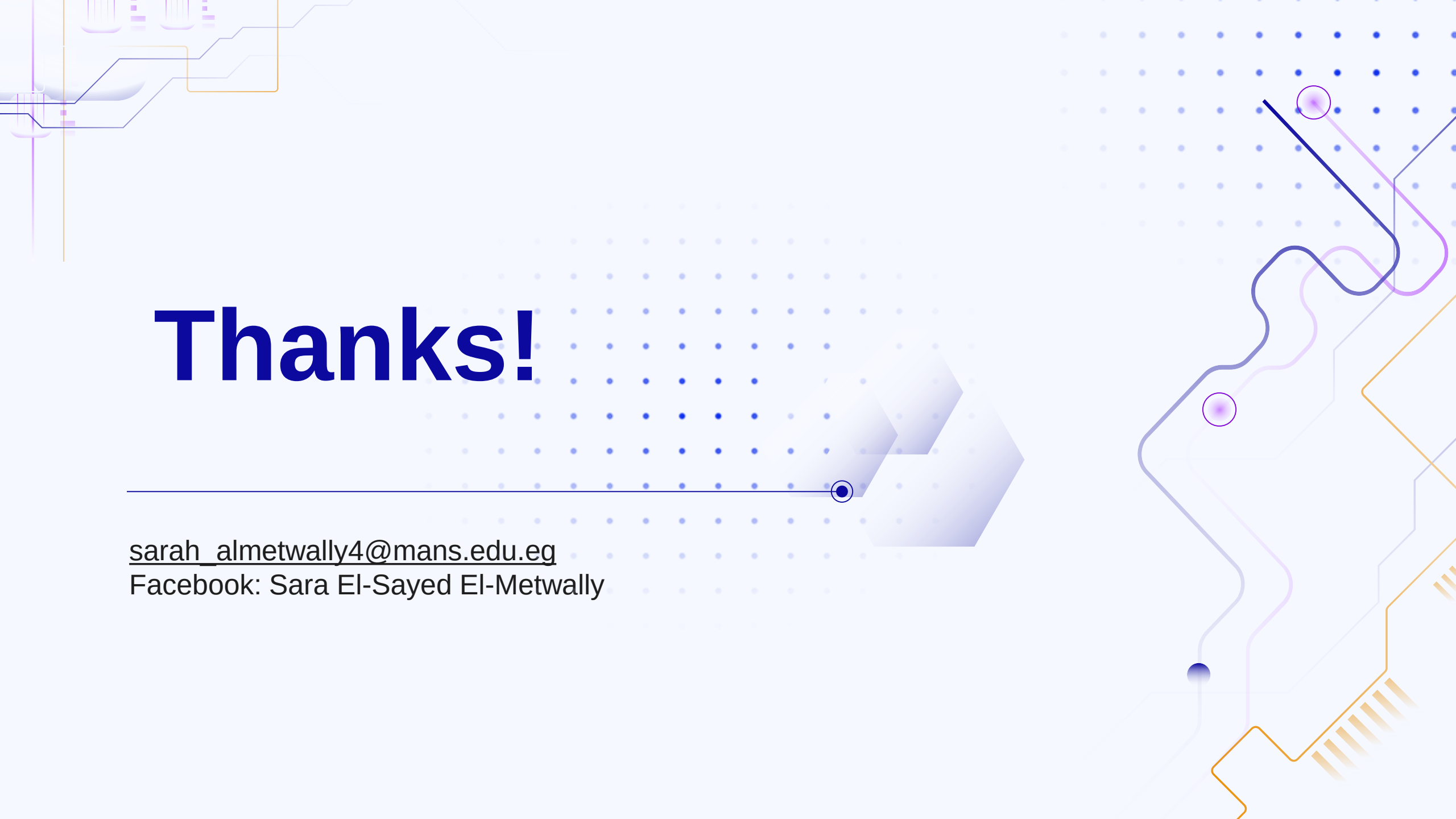
# Load the image into the array
data[0] = normalized_image_array
```

Google Teachable Machine: Model Export

```
# Predicts the model
prediction = model.predict(data)
index = np.argmax(prediction)
class_name = class_names[index]
confidence_score = prediction[0][index]

# Print prediction and confidence score
print("Class:", class_name[2:], end="")
print("Confidence Score:", confidence_score)
```

```
1/1 [=====] - 1s 1s/step
Class: Skin Cancer
Confidence Score: 1.0
```

Thanks!

sarah_almetwally4@mans.edu.eg

Facebook: Sara El-Sayed El-Metwally